

Assignment No- 02 Design of Machine Element against static loading

Q.1) State & explain

A) Max. normal stress theory / Max principle stress theory

B) Max. shear stress theory

C) Max. distortion energy theory

→ A) Max. normal stress theory / Max. principle stress theory.

- This criterion of failure is accredited to the British engineer W.J.M. Rankine (1850). The theory states that the failure of the mechanical component subjected to bi-axial or tri-axial stresses occurs when the maximum principle stress reaches the yield or ultimate strength of the material.

- If σ_1, σ_2 and σ_3 are the three principal stresses at a point on the component and

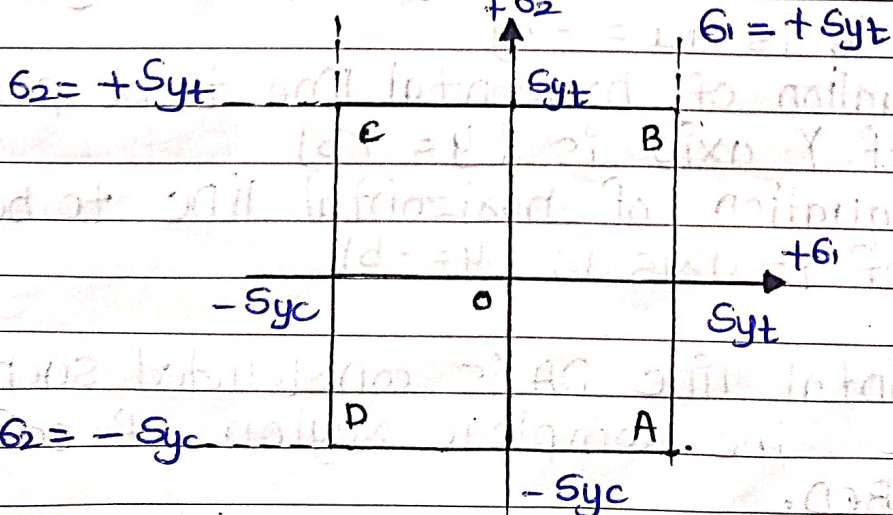
$$\sigma_1 > \sigma_2 > \sigma_3$$

- then according to this theory the failure occurs whenever

$$\sigma_1 = \sigma_{yt} \text{ or } \sigma_1 = \sigma_{ut}$$

- Whichever is applicable

$$\sigma_1 = -\sigma_{yc}$$



- The theory considers only the maximum of principal stresses and disregards the influence of the other principal stresses.
- The dimension of the component are determined by using a factor of safety
- For tensile stresses:

$$\sigma_1 = \frac{\sigma_{yt}}{(f_s)} \quad \text{or} \quad \sigma_1 = \frac{\sigma_{ut}}{(f_s)}$$

- For compressive stresses

$$\sigma_1 = \frac{\sigma_{yc}}{(f_s)} \quad \text{or} \quad \sigma_1 = \frac{\sigma_{sc}}{(f_s)}$$

Region of safety: The construction of region of safety for bi-axial stresses is illustrated in fig.

- The two principal stresses σ_1 & σ_2 are plotted on X and Y axes respectively.

- Tensile stresses are considered as positive, while compressive stresses as negative. It is further assumed that.

$$\sigma_{yc} = \sigma_{yt}$$

- It should be noted that

- The equation of vertical line to be positive side of X-axis is $(x = +a)$
- The equation of vertical line to be negative side of X-axis is $(x = -a)$
- The equation of horizontal line to be positive side of Y-axis is $(y = +b)$
- The equation of horizontal line to be negative side of Y-axis is $(y = -b)$

- A horizontal line DA is constructed such that $\sigma_2 = -\sigma_{yc}$. The complete region of safety is area ABCD.

- Since we have assumed ($S_{yc} = S_{yt}$), ABCD is square.
- According to the maximum principal stress theory of failure if a point with co-ordinates (σ_1, σ_2) fall outside this square, then it indicates the failure condition.
- On the other hand if the point falls inside the square the design is safe and the failure mode may not occur.
- Experimental investigations suggest that the maximum principal stress theory gives good predictions for brittle materials. However it is not recommended for ductile materials.

B] Max. Shear stress theory

- This criterion of failure is accredited to C.A. Coulomb, H. Tresca and J.J. Guest. The theory states that the failure of a mechanical component subjected to bi-axial or tri-axial stresses occurs when the maximum shear stress at any point in the component becomes equal to the maximum shear stress in the standard specimen of the tension test, when yielding starts.
- In tension test, the specimen is subjected to uni-axial stress (σ_1) and ($\sigma_2 = 0$).
- The stress in the specimen of tension test and the corresponding Mohr's circle diagram are as shown in fig.

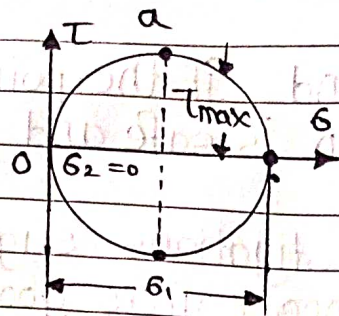
$$\tau_{max} = \frac{\sigma_1}{2}$$

- When the specimen starts yielding ($\sigma_1 = S_{yt}$) the above equation is written as

$$\tau_{max} = \frac{S_{yt}}{2}$$

- Therefore, the maximum shear stress theory predicts that the yield strength in shear is half of the yield strength in tension

$$S_{sy} = 0.5 S_{yt}$$



d) Stresses in Simple tension test

- The box-deline for region of safety for this theory can be constructed in the following way:

Step-1:

- In the first quadrant both (σ_1) & (σ_2) are positive or tensile stresses. Suppose $\sigma_1 > \sigma_2$

- The boundary line will be

$$\sigma_1 = \sigma_{yt}$$

- A vertical line AB is constructed such that $\sigma_1 = +\sigma_{yt}$

Suppose $\sigma_2 > \sigma_1$

- The boundary line will be

$$\sigma_2 = \sigma_{yt}$$

Step-2:

- In the third quadrant both (σ_1) and (σ_2) are negative or compressive stresses

- Suppose $\sigma_1 > \sigma_2$ the boundary line will be $\sigma_2 = -\sigma_{yt}$

- A vertical line DE is constructed such that $\sigma_1 = -\sigma_{yt}$

- Suppose $\sigma_2 > \sigma_1$ the boundary line will be $\sigma_2 = -\sigma_{yt}$

- A horizontal line EF is constructed such that $\sigma_2 = -\sigma_{yt}$

Step-3:

- In second and fourth quadrants (σ_1) and (σ_2) are of opposite sign.

- The yielding will occur when.

$$\sigma_1 = \sigma_2 = \pm \sigma_{yt}$$

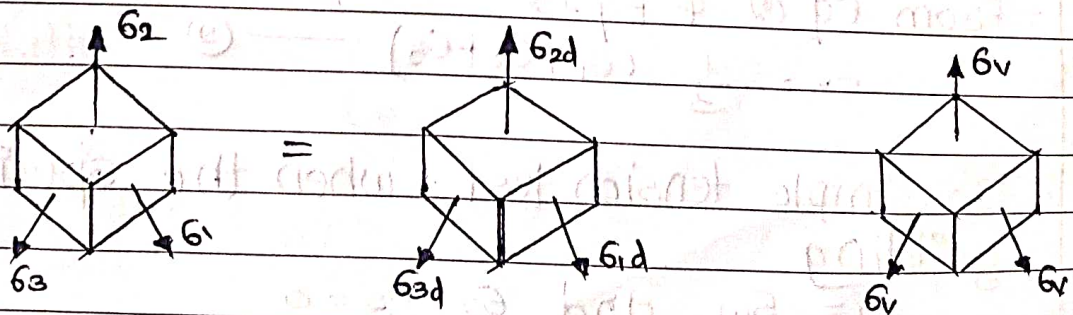
- In second quadrant, line $\overline{DC}$ is constructed such that $\sigma_1 - \sigma_2 = -\sigma_{yt}$
- It is observed that intercept of above line on X-axis and intercept on Y-axis.

Step - 4 :

- In the fourth quadrant, line $\overline{FA}$ is constructed such that $\sigma_1 - \sigma_2 = +\sigma_{yt}$
- It is observed that intercept of above line on X-axis ($\sigma_2 = 0$) is $(+\sigma_{yt})$ and intercept on Y-axis ($\sigma_1 = 0$) is $(-\sigma_{yt})$.
- The complete region of safety is hexagon $ABCBFA$.
- The maximum shear stress theory of failure is widely used by designers for predicting the failure of components, which are made of ductile materials like transmission shaft.

□ Distortion - Energy Theory :

- This theory was advanced by M.T. Huber in Poland (1904) and independently by R. von Mises in Germany (1913) and H. Hencky (1925)
- It is known as the Huber von Mises and Hencky's theory.
- A unit cube subjected to the three principal stresses σ_1, σ_2 and σ_3 is shown in fig.
- The total strain energy U of the cube is given by.



□ Element with tri-axial stresses

- stress components due to distortion of element
- stress components due to change of volume.

$$U = \frac{1}{2} \sigma_1 \epsilon_1 + \frac{1}{2} \sigma_2 \epsilon_2 + \frac{1}{2} \sigma_3 \epsilon_3 \quad \text{--- (1)}$$

where ϵ_1, ϵ_2 & ϵ_3 are strain in respective direction

$$\text{Also, } \epsilon_1 = \frac{1}{E} [\sigma_1 - \mu (\sigma_2 + \sigma_3)]$$

$$\epsilon_2 = \frac{1}{E} [\sigma_2 - \mu (\sigma_1 + \sigma_3)]$$

$$\epsilon_3 = \frac{1}{E} [\sigma_3 - \mu (\sigma_1 + \sigma_2)] \quad \text{--- (2)}$$

- Substituting the above expression in equation (1)

$$U = \frac{1}{2E} [\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\mu (\sigma_1 \sigma_2 + \sigma_2 \sigma_3 + \sigma_3 \sigma_1)] \quad \text{--- (3)}$$

- Therefore,

$$U = U_v + U_d \quad \text{--- (4)}$$

- The corresponding stresses are also resolved into two components as shown in fig.

$$\text{- From the fig. } \sigma_1 = \sigma_{1d} + \sigma_v$$

$$\sigma_2 = \sigma_{2d} + \sigma_v$$

$$\sigma_3 = \sigma_{3d} + \sigma_v \quad \text{--- (5)}$$

- Since the component σ_{1d}, σ_{2d} and σ_{3d} do not change the volume of the cube.

$$\epsilon_{1d} + \epsilon_{2d} + \epsilon_{3d} = 0 \quad \text{--- (6)}$$

- Also,

$$\epsilon_{1d} = \frac{1}{E} [\sigma_{1d} - \mu (\sigma_{2d} + \sigma_{3d})]$$

$$\epsilon_{2d} = \frac{1}{E} [\sigma_{2d} - \mu (\sigma_{1d} + \sigma_{3d})]$$

$$\epsilon_{3d} = \frac{1}{E} [\sigma_{3d} - \mu (\sigma_{1d} + \sigma_{2d})] \quad \text{--- (7)}$$

- From Eq. (7) & (6)

$$\therefore (\sigma_{1d} + \sigma_{2d} + \sigma_{3d}) = 0 \quad \text{--- (8)}$$

- From eq (3) & Eq. (7)

$$\sigma_v = \frac{1}{3} (\sigma_1 + \sigma_2 + \sigma_3) \quad \text{--- (9)}$$

- In simple tension test, when the specimen starts yielding

$$\sigma_1 = \sigma_{yt} \text{ and } \sigma_2 = \sigma_3 = 0$$

- Therefore,

$$U_d = \left(\frac{1+\mu}{3E} \right) \sigma_{yt}^2$$

- The criterion of failure for the distortion energy theory is expressed as.

$$\sigma_{yt}^2 = [(s_1 - s_2)^2 + (s_2 - s_3)^2 + (s_3 - s_1)^2]$$

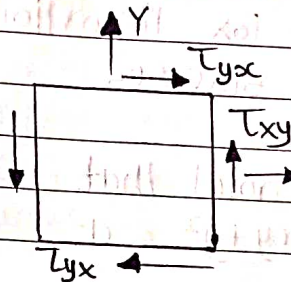
$$\text{or } \sigma_{yt} = \sqrt{\frac{1}{2} [(s_1 - s_2)^2 + (s_2 - s_3)^2 + (s_3 - s_1)^2]} \quad (10)$$

- Considering factor of safety

$$\frac{\sigma_{yt}}{F_s} = \sqrt{\frac{1}{2} [(s_1 - s_2)^2 + (s_2 - s_3)^2 + (s_3 - s_1)^2]}$$

- For bi-axial stress $s_3 = 0$

- A component subjected to pure shear stresses and the corresponding Mohr's circle diagram is shown in fig.



- Element subjected to pure shear stresses

- Mohr's Circle for shear stresses

- From the figure,

$$s_1 = -s_2 = T_{xy} \text{ and } s_3 = 0$$

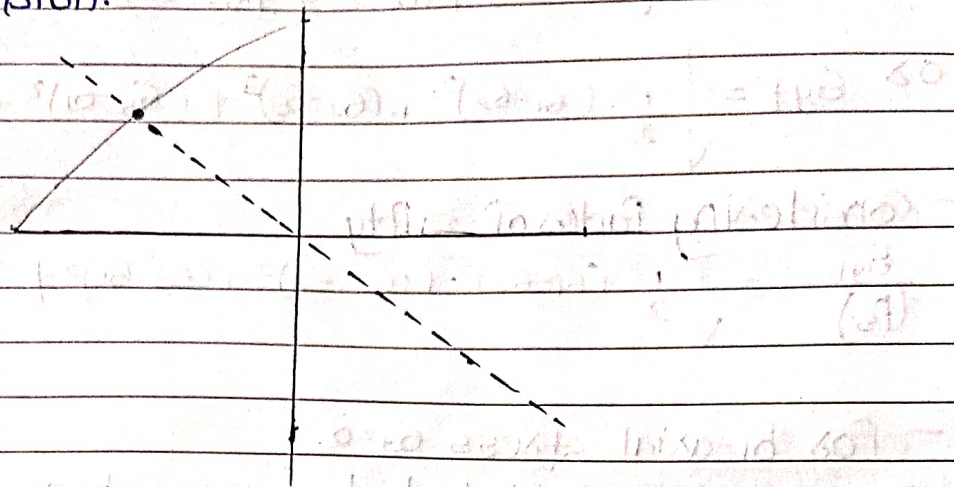
- Substituting these value in equation (10)

$$\sigma_{yt} = \sqrt{3} T_{xy}$$

- Replacing (T_{xy}) by S_{sy}

$$S_{sy} = \frac{\sigma_{yt}}{\sqrt{3}} = 0.577 \sigma_{yt}$$

- Therefore according to the distortion energy theory the yield strength in shear is 0.577 times the yield strength in tension.



- fig. Boundary for distortion energy theory under bi-axial stresses

- It should be noted that

$$x^2 + xy + y^2 = a^2$$

- Is equation of an ellipse whose semi-major axis is $(\sqrt{2}a)$ and semi-minor axis is $\sqrt{2/3}a$

- For bi-axial stresses

$$\sigma_3 = 0$$

- Substituting this value in equation (10)

$$\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2 = S_y^2$$

- Q.2] Design process of knuckle joint.
- The basic procedure to determine the dimension of the knuckle joint consists of following steps.

1] Calculate the diameter of each rod by Eq (4.40.a)

$$D = \sqrt{\frac{4P}{\pi \sigma_t}}$$

2] Calculate the enlarged diameter of each rod by empirical relationship using Equation (4.40 b)

$$D_1 = 1.1 D$$

3] Calculate the dimensions a and b by empirical relationship using equation (4.40 m) and (4.40 n)

$$a = 0.75 D \quad \text{and} \quad b = 1.25 D$$

4] Calculate the diameter of the pin by shear consideration using Equation and bending consideration using equation and select the diameter, whichever is maximum,

$$d = \sqrt{\frac{2P}{\pi \tau}} \quad \text{or} \quad d = \sqrt[3]{\frac{32}{\pi 6b} \times \frac{P}{2} \left[\frac{b}{4} + \frac{a}{3} \right]}$$

5] Calculate the dimensions d_o and d_i by empirical relationship using equation and respectively.

$$d_o = 2d \quad \text{and} \quad d_i = 1.5d$$

6] Check the tensile crushing and shear stresses in the eye by equation (4.40 h) and equation (4.40 i) resp.

$$\sigma_c = \frac{P}{b(d_o - d)}$$

$$\sigma_t = \frac{P}{bd} \quad ; \quad \tau = \frac{P}{b(d_o - d)}$$

7] Check the tensile, crushing and shear stresses in the fork by equation (4.40 k), (4.40 f) and (4.40 l) respectively.

$$G_1 = \frac{P}{2a(d_o - d)}$$

$$G_2 = \frac{P}{2ad}; \quad t = \frac{P}{2a(d_o - d)}$$

- The application of the above mentioned procedure is illustrated in the next example.

Q.3] Design process of cotter joint.

- The basic procedure to calculate the dimensions of the cotter joint consists of following steps:

1] Calculate the diameter of each rod by eqⁿ (4.37 a)

$$d = \sqrt{\frac{4P}{\pi G_1}}$$

2] Calculate the thickness of the energy cotter by empirical relationship given in equation (4.37 c)

$$t = 0.31d$$

3] Calculate the diameter d_2 of the spigot on the basis of tensile stress. From equation (4.37 b)

$$P = \left[\frac{\pi}{4} d_2^2 - d_2 t \right] G_1$$

- When the values of P , t and G_1 are substituted the above expression becomes a quadratic equation.

4] Calculate the outside diameter d_1 of the socket on the basis of tensile stress in the socket from eq (4.37 d)

$$P = \left[\frac{\pi}{4} (d_1^2 - d_2^2) - (d_1 - d_2)t \right] G_1$$

- When the value of P , d_2 , t and G_1 are substituted the above expression becomes a quadratic equation

- 5] The diameter of the spigot collar d_3 and the diameter of the socket collar d_4 are calculated by following empirical relationships.

$$d_3 = 1.5d, \quad d_4 = 2.4d$$

- 6] The dimensions a and c are calculated by following empirical relationships.

$$a = c = 0.75d$$

- 7] Calculate the width b of the collar by shear consideration using equation (4.37c) and bending consideration using equation (4.37j) and select the width whichever is maximum between these two values

$$b = \frac{P}{2\tau t} \text{ or } \sqrt{\frac{3P}{t\sigma_b} \left[\frac{d_2 + d_4 - d_2}{4} \right]}$$

- 8] Check the crushing and shear stresses in the spigot end by equation (4.37h) and (4.37-f) resp.

$$\sigma_c = \frac{P}{td_2}, \quad \tau = \frac{P}{2ad_2}$$

- 9] Check the crushing and shear stresses in the select the socket end by equation (4.37j) and (4.37g) respectively.

$$\sigma_c = \frac{P}{(d_4 - d_2)t}; \quad \tau = \frac{P}{2(d_4 - d_2)c}$$

- 10] Calculate the thickness t_1 of the spigot collar by following empirical relationship

$$t_1 = 0.45d$$

-The taper of the collar is 1 in 32